

Tabular Q-Learning on FrozenLake-v1

Critical Analysis Report — Yonatan Azmir, June 2026

Overview

A tabular Q-learning agent was implemented from scratch using Python and NumPy to solve the FrozenLake-v1 environment under two regimes: a deterministic **Non-Slippery** setting and a stochastic **Slippery** setting, where actions succeed as intended only one third of the time. The agent learns a 16-state x 4-action Q-table via the Bellman update rule, with an epsilon-greedy policy that decays over training to shift from exploration toward exploitation.

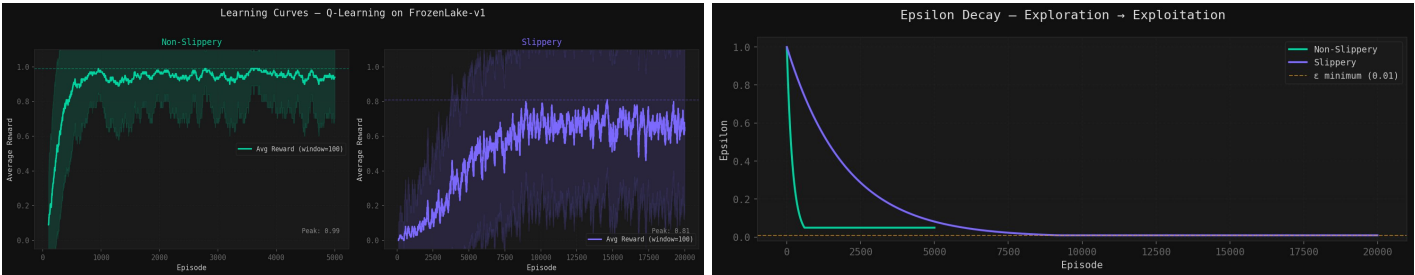
Hyperparameters

Parameter	Non-Slippery	Slippery	Rationale
Episodes	5,000	20,000	Stochastic dynamics require far more samples to converge
alpha (learning rate)	higher	lower	Lower alpha smooths noisy updates from random slips
gamma (discount)	standard	higher	Longer horizon needed to plan around slip risk
epsilon decay	fast (~500 ep)	slow (~9,000 ep)	Slippery env needs prolonged exploration to map outcomes
epsilon end	0.05	0.01	Final exploitation floor

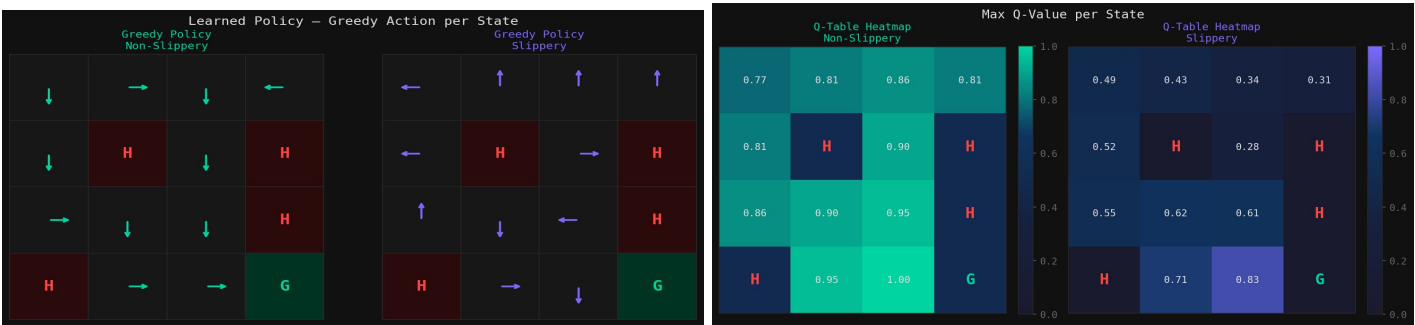
Results (Final Evaluation, 100 Greedy Episodes)

Metric	Non-Slippery	Slippery
Success rate	100 / 100 (100%)	77 / 100 (77%)
Avg. reward (last 1000 train ep)	0.945	0.673
Peak training avg reward	0.99	0.81
Final epsilon	0.0500	0.0100

Learning Dynamics



Learned Policy and State Values



Critical Analysis

Convergence speed. The Non-Slippery agent converges almost immediately, reaching its peak average reward (0.99) within roughly 1,000 episodes and maintaining a near-perfect 100% success rate thereafter. The Slippery agent converges far more slowly and noisily, requiring approximately 9,000-10,000 episodes to plateau around 0.65-0.70 average reward, with a final evaluation success rate of 77%.

Why the gap exists. Under stochastic transitions, the same (state, action) pair can lead to different outcomes roughly two-thirds of the time, so a single Bellman update is a noisy sample of the true transition distribution. This forces the learned policy toward routes that are robust to unintended slips rather than the shortest path, and it caps the achievable success rate well below 100% even with a near-optimal policy, since some slips are simply unrecoverable. The Q-table heatmap reflects this: Non-Slippery values increase monotonically toward the goal (reaching 1.00), while Slippery values are lower and less smooth, indicating the agent has learned to hedge against risk rather than commit to a single deterministic trajectory.

Exploration-exploitation schedule. The epsilon decay plot shows the Non-Slippery agent exhausts exploration in roughly 500 episodes, appropriate for a small, deterministic state space with little ambiguity. The Slippery agent decays over ~9,000 episodes, which the state-visitation heatmap confirms was necessary: visitation counts there are orders of magnitude higher and far more spread across non-terminal states, showing the agent needed sustained exploration to adequately sample the noisy outcomes of each action before committing to a policy.

Policy quality. The greedy policy arrows show the Non-Slippery agent finds a direct, efficient path to the goal with no wasted moves. The Slippery agent's policy instead favors actions that push the agent away from holes even when slipping (e.g., aiming 'into' a wall to reduce the chance of a slip carrying it into a hole), a classic risk-averse adaptation to stochastic dynamics.

Limitations. The tabular approach scales only to small, discrete state spaces and is sensitive to hyperparameter choices, particularly the epsilon decay rate and learning rate α under stochasticity. The 77% success rate, while strong, also reflects an inherent ceiling imposed by the environment's randomness rather than a deficiency in the learned policy itself.